KHUSHBOO

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PROFILE

Robotics & Artificial Intelligence engineering student at Thapar Institute of Engineering and Technology, graduating in 2027, with hands-on experience in Machine Learning, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Python, data analysis, and AI application development. Built a multi-agent RAG platform with retrieval evaluation, vector search, BM25, LangGraph, and Streamlit, along with projects involving SQL optimization, data analytics, and autonomous robotics. Strong foundation in AI/ML experimentation, data processing, model evaluation, problem-solving, and developing practical AI solutions.

EDUCATION

Thapar Institute of Engineering and Technology, Patiala, Punjab 2023 – 2027

B.E., Robotics & Artificial Intelligence — CGPA: 7.28/10 (till 6th Sem)

Triple M Public School, Hoshiarpur

High School — PCM Percentage: 90%

WORK EXPERIENCE

Deloitte | Data Analytics Virtual Experience Program (Forage) September 2026

- Completed a **Deloitte Data Analytics Job Simulation** on Forage, applying practical skills in data analysis and forensic technology to a simulated client engagement.
- Analyzed factory telemetry data using **Tableau** to identify machine downtime patterns, high-risk locations, and frequently failing device types, surfacing actionable operational insights.
- Built an interactive **Tableau dashboard** using calculated fields, bar charts, and dashboard filters to visualize KPIs and support data-driven decision-making.
- Performed data classification and analysis on employee compensation data using Excel to assess gender pay equality across factories and job roles, and summarized findings for stakeholder review.

TECHNICAL SKILLS

AI / Machine Learning: Machine Learning, Predictive Analytics, Deep Learning, Generative AI, Large Language Models (LLMs), AI Agents, Multi-Agent Systems, Explainable AI, Model Evaluation, scikit-learn, TensorFlow, PyTorch

LLM / RAG: Retrieval-Augmented Generation (RAG), Vector Search, BM25, Reranking, LangGraph, Information Retrieval, Document Chunking, Retrieval Evaluation

Programming: Python, SQL, C, MATLAB, Data Structures & Algorithms, Object-Oriented Programming

Data & Analytics: Data Preprocessing, Exploratory Data Analysis, Business Intelligence, Data Analysis, Statistical Analysis, KPI Analysis, Data Visualization

Databases & Query Optimization: PostgreSQL, Query Optimization, EXPLAIN ANALYZE, Database Indexing, Composite Indexing, Materialized Views, DBMS

AI / Data Applications: Streamlit, Dashboard Development, Tableau, Microsoft Excel, Microsoft PowerPoint

Tools & Platforms: Git, GitHub, Linux, Google Cloud Platform (GCP)

Robotics & Simulation: ROS2, SLAM, Sensor Fusion, Path Planning, Gazebo, URDF, RViz

PROJECTS

Multi-Agent AI Research & Decision Intelligence Platform [GitHub] | *Python, LangGraph, RAG, BM25, Vector Search, Streamlit*

- Developed an offline, evidence-grounded RAG platform orchestrating six specialized agents for research, data analysis, calculation, synthesis, critique, and claim verification.
- Indexed 7 heterogeneous documents into 65 chunks and evaluated retrieval accuracy and system effectiveness, achieving 0.792 precision@k, 0.833 recall@k/MRR, 100% citation and claim-support coverage, 83.62% confidence, and 4.3 ms average latency across 6 labelled evaluation queries.
- Added an evaluation dashboard and system tracing/observability to continuously test and validate answer quality, plus a Streamlit interface for stakeholder interaction.

E-commerce Order Intelligence & SQL Performance Optimization [GitHub] | *PostgreSQL, EXPLAIN ANALYZE, Streamlit*

- Built and analyzed a PostgreSQL marketplace dataset containing 50K orders and 100K order items, translating business requirements around seller delivery, cancellations, and revenue leakage into SQL-based analytical questions.
- Used EXPLAIN ANALYZE, composite indexing, sargable filters, and a materialized KPI layer to reduce seller-hotspot query runtime by 82.6%, from 66.7 ms to 11.6 ms.
- Developed a Streamlit decision dashboard to communicate operational KPIs and insights, supporting faster data-driven decision-making.

Autonomous Indoor Delivery Robot [GitHub] | *Python, ROS2, Gazebo, URDF, SLAM, Sensor Fusion, RViz2*

- Modeled a four-wheel delivery robot in Creo/SolidWorks, converted it to a URDF-based ROS2 simulation, and integrated it into a custom hospital environment built in Gazebo Ignition Fortress.
- Fused LiDAR, camera, and IMU data for stable real-time localization in GPS-denied indoor spaces, and used SLAM to generate a testable occupancy-grid map visualized in RViz2.
- Iteratively debugged and tuned navigation and obstacle-avoidance logic against real-world performance data, reducing path-planning failures by ~30% versus baseline.

HACKATHONS, ACHIEVEMENTS & LEADERSHIP

- TVS Credit E.P.I.C 8.0 — IT Challenge: Participant.
- Adobe University Hackathon 2026 — Team Leader.
- Scored 100% in Class XII Mathematics.
- Core Member, **MECIVIONS Research Society**, Thapar Institute — contributed to research initiatives and technical activities.

CERTIFICATIONS

Data Science (Udemy, 2025) | Machine Learning (Udemy, 2025) | Python Programming (Udemy, 2025) | ROBO AI CARLO (my equation, 2026)